AI ACADEMY · LEVEL 6 · GRADE 11 · WEEK 5 · WORKBOOK

Authentication and authorisation

Learn to understand, design, build, test and explain AI systems responsibly.

Project: Grounded Learning Service

THE CONNECTED WEEK

One complete authentication-and-permission reasoning record.

Keep trusted context, request claims, action, object, decision and evidence limits together.

Keep one habit: Recall → W1–W3/W5 → individual role check → project permission policy → delayed return.

MAIN LESSON · SECTION 1

Recall · Retrieve before notes

> Keep this task's inputs,rule and response together.

Complete task and evidence

- Try the recall before opening notes. Then use a different colour or a labelled second line to correct your explanation.

- Without opening the old lesson, explain one key idea from Configuration and secrets. Give an example and a mistake that the idea helps you avoid.

- Without opening the old lesson, explain one key idea from Architecture and data flow. Give an example and a mistake that the idea helps you avoid.

- Use this reference for the original W1–W3 and W5 exercises. The separately named independent case supplies its own changed facts; use those facts for that attempt before feedback.

- **Authentication** identifies an actor; **authorisation** decides whether that actor may perform a particular operation on a particular resource.

- Use trusted session roles, never a role claimed inside user JSON. Student may read student resources; teacher may read both. Unauthenticated requests are denied.

Configuration and secrets

> **My remembered idea, example and correction**

Architecture and data flow

> **My remembered idea, example and correction**

Lesson reference for workbook practice

W1 · Explain the current example

> Keep this task's inputs,rule and response together.

Complete task and evidence

- School service policy: trusted student may read student resources; trusted teacher may read student and teacher resources; no recognized trusted session or unknown role/resource classification denies. Main request comes from a trusted student session but carries role=teacher and asks to read a teacher guide. Use trusted session role, deny before reading guide body, and leave session unchanged.

  Client-written roles never promote authority. Authentication establishes identity/session context; authorization checks action/resource permission using trusted policy. Local code starts with trusted fixtures, not a real login. Separate museum policy: verified volunteer Jo may read C but not edit C; verified curator with edit permission passes that gate, subject to remaining checks.

  Permission is not executed editing. No real authentication protocol, expiry, writes or deployment are implemented. Later school role decisions apply the already taught policy; collect individual reasoning and reserve a fresh action/resource case for new-transfer claims.

- Explain the weekly **mechanism** in your own words. Use the worked scenario to name its **input**, decision rule and **output**.

> **My reasoning and evidence**

W2 · Trace the calculation

> Keep this task’s inputs,rule and response together.

Complete task and evidence

- School service policy: trusted student may read student resources; trusted teacher may read student and teacher resources; no recognized trusted session or unknown role/resource classification denies. Main request comes from a trusted student session but carries role=teacher and asks to read a teacher guide. Use trusted session role, deny before reading guide body, and leave session unchanged.

  Client-written roles never promote authority. Authentication establishes identity/session context; authorization checks action/resource permission using trusted policy. Local code starts with trusted fixtures, not a real login. Separate museum policy: verified volunteer Jo may read C but not edit C; verified curator with edit permission passes that gate, subject to remaining checks.

  Permission is not executed editing. No real authentication protocol, expiry, writes or deployment are implemented. Later school role decisions apply the already taught policy; collect individual reasoning and reserve a fresh action/resource case for new-transfer claims.

- A student session submits role=teacher in request data and asks for a guide. Show the intermediate steps and give the conclusion with its scope.

My reasoning and evidence

W3 · Repair a misconception

> Keep this task’s inputs,rule and response together.

Complete task and evidence

- School service policy: trusted student may read student resources; trusted teacher may read student and teacher resources; no recognized trusted session or unknown role/resource classification denies. Main request comes from a trusted student session but carries role=teacher and asks to read a teacher guide. Use trusted session role, deny before reading guide body, and leave session unchanged.

  Client-written roles never promote authority. Authentication establishes identity/session context; authorization checks action/resource permission using trusted policy. Local code starts with trusted fixtures, not a real login. Separate museum policy: verified volunteer Jo may read C but not edit C; verified curator with edit permission passes that gate, subject to remaining checks.

  Permission is not executed editing. No real authentication protocol, expiry, writes or deployment are implemented. Later school role decisions apply the already taught policy; collect individual reasoning and reserve a fresh action/resource case for new-transfer claims.

- A report omits this qualification: Successful **authentication** does not grant every resource or operation. Explain how omitting it could change a reader's interpretation, and write an accurate replacement.

> **My reasoning and evidence**

W5 · Explain the limitation

> Keep this task's inputs,rule and response together.

Complete task and evidence

- School service policy: trusted student may read student resources; trusted teacher may read student and teacher resources; no recognized trusted session or unknown role/resource classification denies. Main request comes from a trusted student session but carries role=teacher and asks to read a teacher guide. Use trusted session role, deny before reading guide body, and leave session unchanged.

  Client-written roles never promote authority. Authentication establishes identity/session context; authorization checks action/resource permission using trusted policy. Local code starts with trusted fixtures, not a real login. Separate museum policy: verified volunteer Jo may read C but not edit C; verified curator with edit permission passes that gate, subject to remaining checks.

  Permission is not executed editing. No real authentication protocol, expiry, writes or deployment are implemented. Later school role decisions apply the already taught policy; collect individual reasoning and reserve a fresh action/resource case for new-transfer claims.

- For the worked scenario, propose one additional case that could reveal a failure, describe its expected outcome before running it, and explain what would need to remain fixed for a fair comparison.

  Do not reuse W4 as a new final test after receiving feedback.

My reasoning and evidence

Try alone · Selected transfer

> Keep this task’s inputs,rule and response together.

Complete task and evidence

- Actor roles are none, student, teacher; target is teacher guide. Decide each under the policy.
- Attempt this case before feedback. Record any help. Earlier examples below are further practice; a case already practised with feedback cannot establish fresh transfer.

Independent case: Apply a new case independently

> **My independent explanation, trace and limits**

MAIN LESSON · SECTION 10

Project · Keep the service evidence

> Preserve the contract, inputs, observed outcomes and limits.

Complete task and evidence

- Add a labelled **authentication** and **authorisation evidence** sheet to Grounded Learning Service. Show how you can explain the **mechanism** and terminology of **authentication** and **authorisation**.

  Use the supplied fictional case and keep its **input**s, intermediate steps, calculation or **prediction**, and limitation. Connect this record to last week's record; a matching answer without a trace is insufficient.

> **My project change, supporting evidence and remaining limit**

MAIN LESSON · SECTION 11

Feedback · Repair and return

> Keep this task's inputs,rule and response together.

Complete task and evidence

- Revisit this idea with your teacher after a delay. Familiarity is different from explaining and using it on a changed case.

Evidence	My record
Worked alone / hint / model	
First step I repaired and why	
New case checked later; date and result	

Evidence	My record
What I can now explain without notes	
One remaining question	

Reference bank and extra practice

Choose the reference that helps your next step.

▼ **Reference · Supported W2 record**

REFERENCE · SECTION 2

Reference · Supported W2 record

> Keep this task's inputs,rule and response together.

Complete task and evidence

- Use W2: Trace the worked case for **Authentication** and **authorisation**. Record the answer once; the lesson and workbook refer to the same attempt.

Given inputs	Procedure I chose	Middle steps	Result and check

The step I completed without help, and the hint I needed

▼ Extra practice · Individual reasoning check

REFERENCE · SECTION 3

Extra practice · Individual reasoning check

> Keep this task's inputs,rule and response together.

Complete task and evidence

- Use the worked case for **Authentication** and **authorisation**. Proposed explanation: "The worked result establishes every future case". Decide whether this explanation is supported.

  Point to the step or supplied fact that decides; explain your repair if needed.

> **My explanation before partner discussion**

▼ Reference · Earlier W4 response copy

REFERENCE · SECTION 8

Reference · Earlier W4 response copy

> Keep this task's inputs,rule and response together.

Complete task and evidence

- Actor roles are none, student, teacher; target is teacher guide. Decide each under the policy.

My reasoning and evidence

▼ **Extra practice · Pictured case**

REFERENCE · SECTION 12

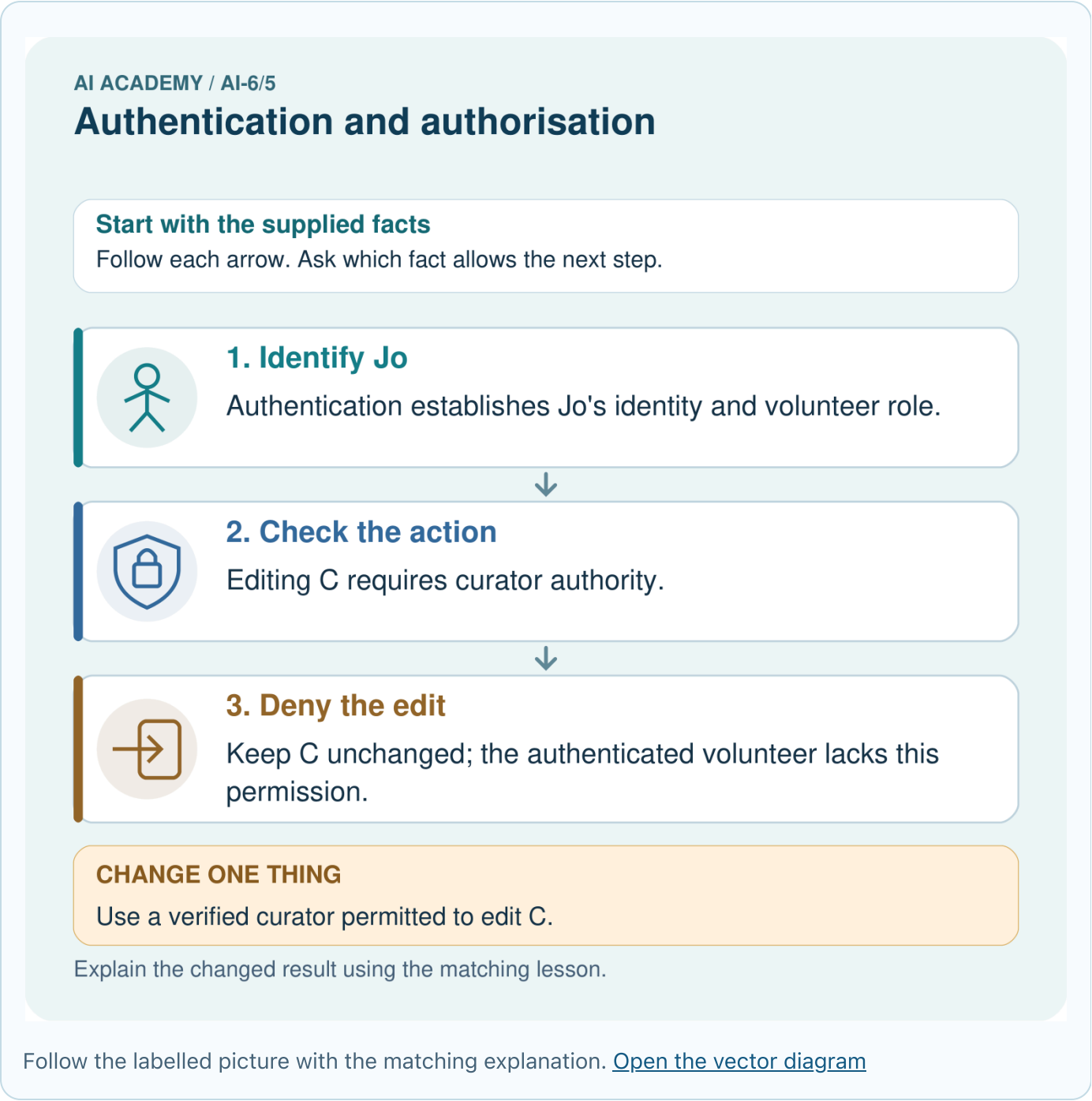
Extra practice · Pictured case

> Keep this task's inputs,rule and response together.

Complete task and evidence

- A verified reader cannot edit. Jo logs in as a verified museum volunteer and asks to edit card C.
- Trusted policy grants volunteers read access to C; only curators may edit it.
- Explain why Jo's valid volunteer login does not permit editing C. Which **trusted role** change allows the action, and why is a role written in the request insufficient?
- Support used: alone / hint / worked example. This record is practice **evidence**.

- Project connection: For the Grounded Learning Service, retain an annotated trace and the changed-case result. Name the contract or **evidence** that makes the deciding step valid, so another team can review it.



Follow the labelled picture with the matching explanation. Open the vector diagram

My picture-based explanation and changed-case prediction

▼ **Reference · Project milestone**

REFERENCE · SECTION 13

Reference · Project milestone

> Keep this task's inputs,rule and response together.

Complete task and evidence

- Project: Grounded Learning Service
- Current milestone: Frame the need. Name the user, the problem, allowed **input**s and a safe simple starting method.
- For the Grounded Learning Service, retain an annotated trace and the changed-case result. Name the contract or **evidence** that makes the deciding step valid, so another team can review it.

Evidence to keep	My record
This week's input and deciding step	
Prediction and actual result (or labelled paper trace)	
One change since the previous checkpoint and why	

Evidence to keep	My record
One error or limit and the next test	

Explain what this week adds to your project. Reuse last week's record rather than starting again.

▼ **Extra practice · Predict,check and repair**

REFERENCE · SECTION 14

Extra practice · Predict,check and repair

> Keep supported feedback separate from the independent case.

First supported question and optional hint

- A verified reader cannot edit. Jo logs in as a verified museum volunteer and asks to edit card C.
- Which labelled step matches this detail? Keep C unchanged; the authenticated volunteer lacks this permission.
- 1. Check the action
- 2. Deny the edit

- 3. Identify Jo
- Hint if needed: Read the steps in the pictorial case. Match the supplied detail to the stage that performs or records it.

Practice 1: explain the deciding evidence

> **My selected conclusion, deciding fact and explanation**

▼ **Check after the first attempt**

After your attempt, compare with the supported explanation: Deny the edit Keep C unchanged; the authenticated volunteer lacks this permission. **Authentication** establishes identity.

Authorisation compares that identity with the requested action and resource.

Changed-case question and optional hint

- Changed-case practice: Use a verified curator permitted to edit C. Which conclusion is supported by the supplied facts?
- 1. We can decide the result without examining the changed **input** or the procedure.
- 2. This one example guarantees correct results for every future **input**.
- 3. The **authorisation** gate allows the edit, subject to other declared checks.
- Hint if needed: Use the changed condition and the deciding step in the pictorial trace. The answer must say what follows from those facts.

Practice 2: explain the deciding evidence

> **My selected conclusion, deciding fact and explanation**
>
> _____
>
> _____
>
> _____
>
> _____

▼ **Check after the changed attempt**

After your attempt, compare with the supported explanation: The **authorisation** gate allows the edit, subject to other declared checks. The **trusted role** now matches this operation and object.

Repair and retain the record

Review Configuration and secrets (ai-6/4). Goal: Explain the **mechanism** and terminology of configuration and secrets. Short practice: Point to the public setting and the protected credential.

What changes when maximum_words becomes 60, and which information must never appear in the startup log?. Return to this week and explain what you changed.

Review Architecture and data flow (ai-6/2). Goal: Explain the **mechanism** and terminology of architecture and data flow. Short practice: Label what moves along the library request, retrieval and drafting arrows.

Where does 17:30 come from, and why must the service stop if the retrieval arrow is disconnected?. Return to this week and explain what you changed.

The browser lab lets you edit the supported Python examples already included in this lesson. Predict first, run, inspect the actual **output** or error, change one **input**, then run again.

Keep code, **prediction**s, actual results and a reasoned revision together. On paper, label your result as a trace rather than an execution.

Use Download completed work in the hosted lesson to export your answers, reflections, practice feedback and code-run **evidence**. Open the HTML copy to read or print it.

Drafts stay in this browser when storage is available; clear drafts on a shared device.

What I repaired, the evidence I kept and the next test

▼ Earlier wording, alternatives and picture transcripts

Use this supported practice once, in the browser or on paper. Keep the reserved independent assessment for a separate first attempt before teacher feedback.

Repair a missing building block

Predict, run and debug

Keep your completed work

▼ Reference · Six-picture response sheets

REFERENCE · SECTION 15

Reference · Six-picture response sheets

> Choose the picture that helps explain the current step.

01. Authentication and authorisation: the case



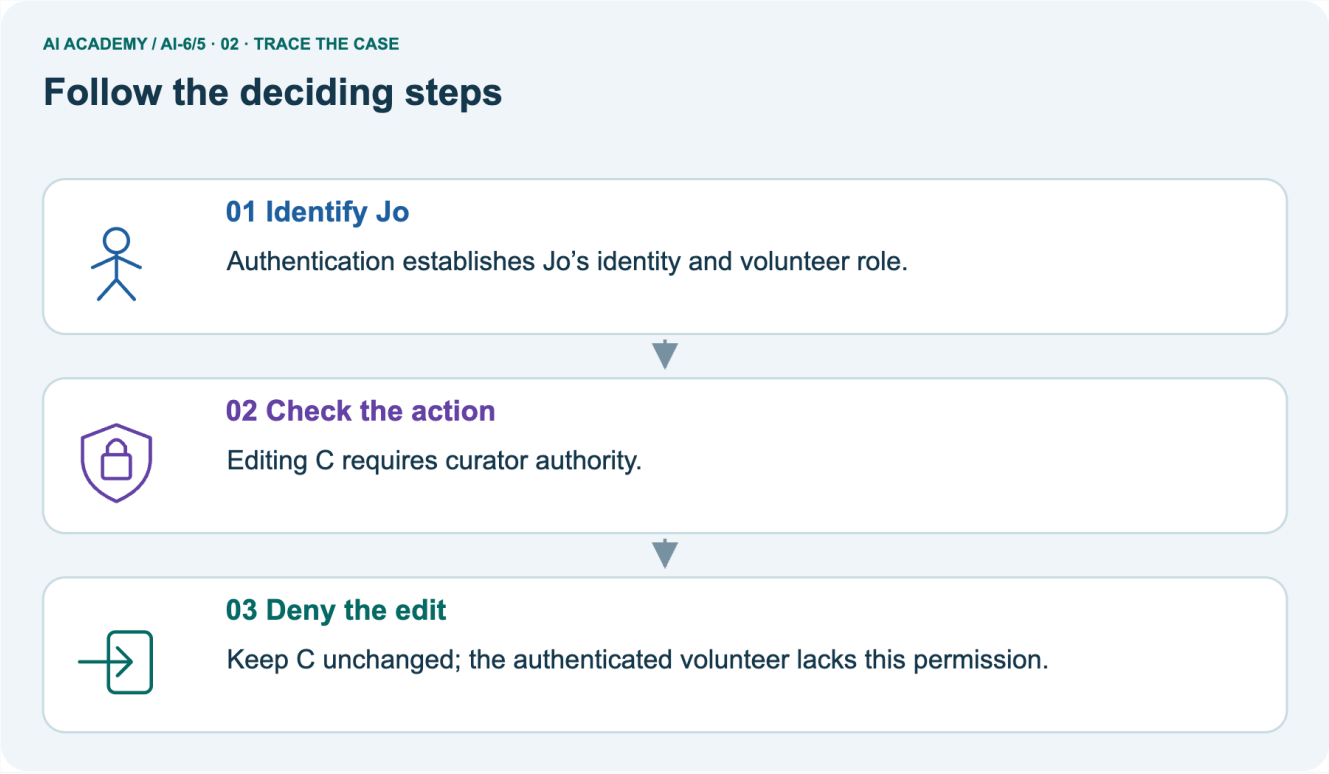
Follow the labelled picture with the matching explanation. Open the vector diagram

A verified reader cannot edit. Jo logs in as a verified museum volunteer and asks to edit card C. Supplied facts: Trusted policy grants volunteers read access to C; only curators may edit it.

Point to the given fact that matters most. Explain why it matters.

My picture explanation, deciding evidence and one limit

02. Follow the deciding steps



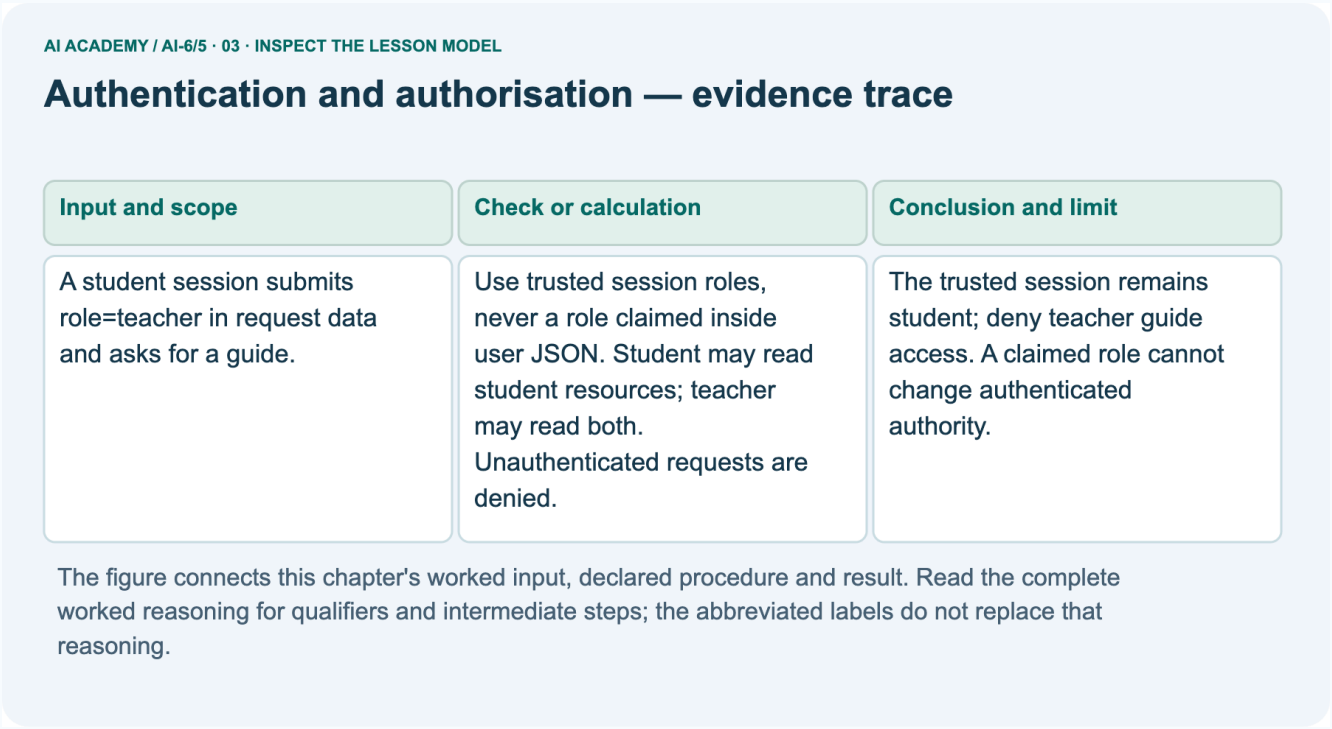
Follow the labelled picture with the matching explanation. Open the vector diagram

Identify Jo: **Authentication** establishes Jo's identity and volunteer role. → Check the action: Editing C requires curator authority. → Deny the edit: Keep C unchanged; the authenticated volunteer lacks this permission.

Point to each numbered stage and explain which recorded **input** it uses.

My picture explanation, deciding evidence and one limit

03. Authentication and authorisation — evidence trace



Follow the labelled picture with the matching explanation. Open the vector diagram

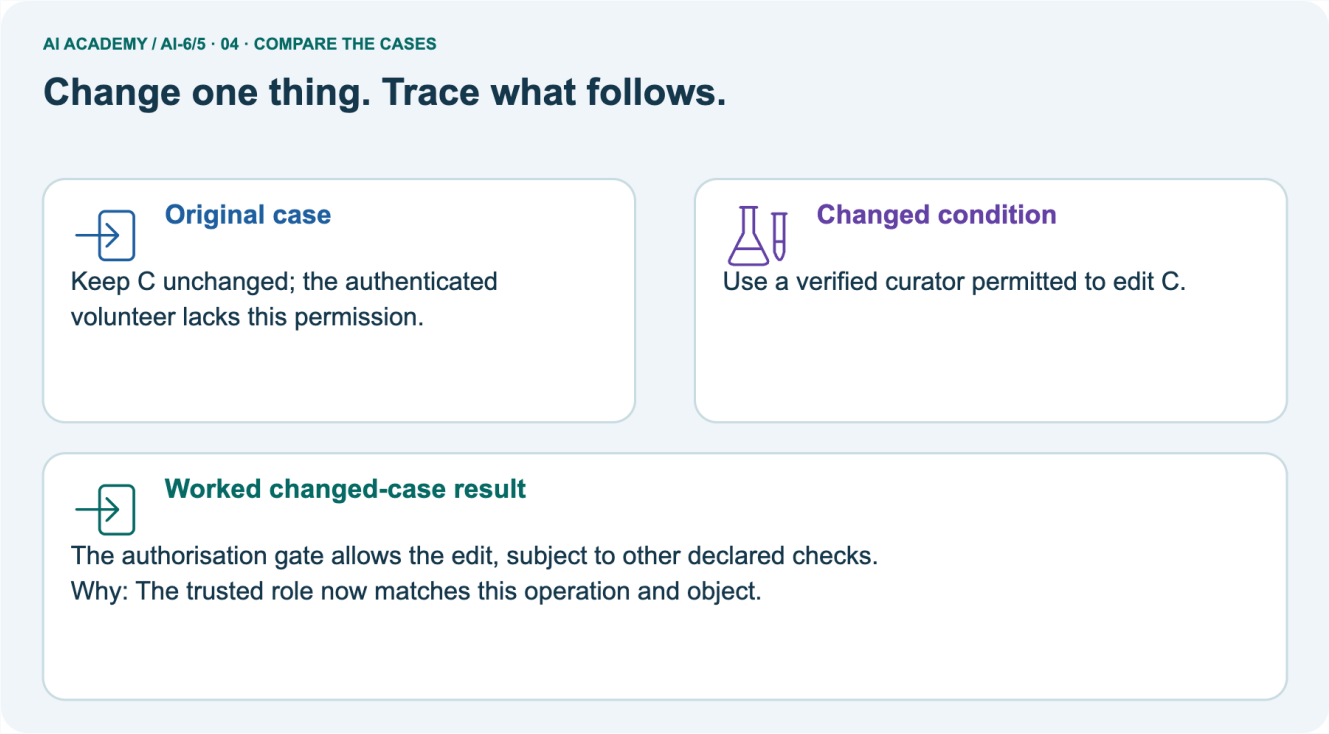
Authentication and **authorisation** — **evidence** trace. A student session submits role=teacher in request data and asks for a guide. | Use trusted session roles, never a role claimed inside user JSON.

Student may read student resources; teacher may read both. Unauthenticated requests are denied. | The trusted session remains student; deny teacher guide access. A claimed role cannot change authenticated authority.

Read the labels and values. Explain the relationship the model shows.

My picture explanation, deciding evidence and one limit

04. Change one thing. Trace what follows.



Follow the labelled picture with the matching explanation. Open the vector diagram

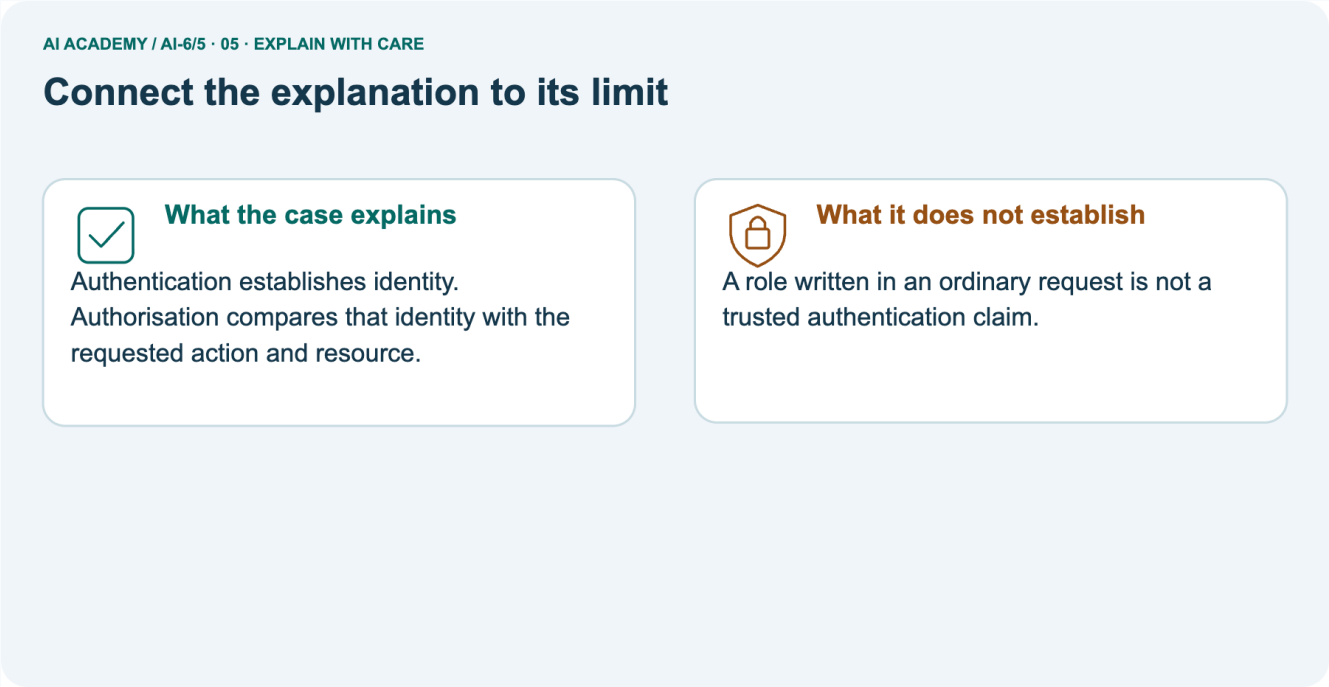
Original case: Keep C unchanged; the authenticated volunteer lacks this permission.
Changed condition: Use a verified curator permitted to edit C.

Expected result: The **authorisation** gate allows the edit, subject to other declared checks.
Reason: The **trusted role** now matches this operation and object.

Predict before checking the worked changed-case result. Name what changed and what the supplied facts allow you to conclude.

My picture explanation, deciding evidence and one limit

05. Connect the explanation to its limit



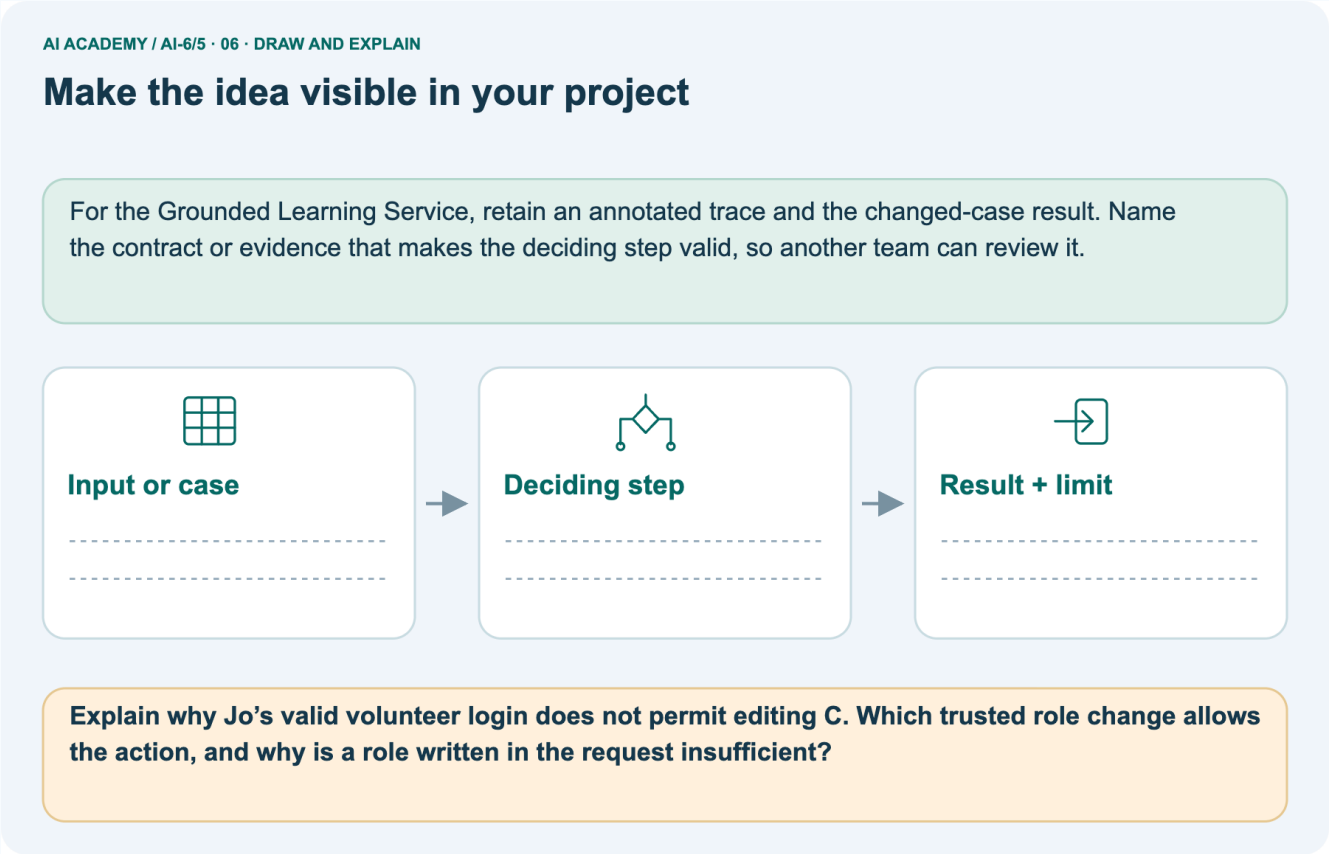
Follow the labelled picture with the matching explanation. Open the vector diagram

Authentication establishes identity. **Authorisation** compares that identity with the requested action and resource. Limit: A role written in an ordinary request is not a trusted **authentication** claim.

Explain the case, then point to a limit on the claim.

My picture explanation, deciding evidence and one limit

06. Make the idea visible in your project



Follow the labelled picture with the matching explanation. Open the vector diagram

For the Grounded Learning Service, retain an annotated trace and the changed-case result. Name the contract or **evidence** that makes the deciding step valid, so another team can review it.

Draw your **input**, deciding step, result and limit. Supported question: Explain why Jo's valid volunteer login does not permit editing C.

Which **trusted role** change allows the action, and why is a role written in the request insufficient?

Draw your project trace and justify the deciding step using **evidence**.

My picture explanation, deciding evidence and one limit

▼ Earlier wording, alternatives and picture transcripts

01. Authentication and authorisation: the case

02. Follow the deciding steps

03. Authentication and authorisation — evidence trace

04. Change one thing. Trace what follows.

05. Connect the explanation to its limit

06. Make the idea visible in your project

▼ Optional interactive practice and completed work

Save or print your responses for your portfolio. Progress and drafts on the hosted website stay in the current browser. Use only fictional or teacher-approved data.